

Introduction to Nyquist-Rate ADCs

As mentioned in Chapter 1, an analog-to-digital converter (ADC) receives a continuous-time, continuous-amplitude signal and transforms it to a discrete-time, discrete-amplitude representation. We also saw that this transformation requires sampling and quantization, and we studied the former in Chapter 3. We devote the remaining chapters of this book to ADC design.

The term “ADC” can be confusing as it refers to the sampler-quantizer cascade in some cases but to the quantizer itself in some other cases. Moreover, it is possible that the same circuit performs sampling and quantization (Chapter 13). Nevertheless, the exact meaning of the term should be clear from the context.

In this chapter, we first examine the concept of quantization more closely. Next, we introduce the performance parameters of ADCs and describe how they are affected by circuit imperfections. Finally, we describe what we mean by “Nyquist-rate ADCs.”

12.1 General Considerations

An ADC generates a digital output, D_{out} , proportional to its analog input, V_{in} :

$$D_{out} \propto V_{in}. \quad (12.1)$$

Since V_{in} has a dimension but D_{out} does not, a proportionality factor is necessary:

$$D_{out} = \alpha V_{in}. \quad (12.2)$$

Note that α has a dimension of 1/voltage. While the input can assume an infinite number of values, the output is selected from a finite number of codes, i.e., the ADC approximates each input level by one of these codes. As was practiced for DACs (Chapter 8), we can construct the input-output characteristic of ADCs. Depicted in Fig. 12.1 for a 3-bit system, the characteristic plots the equivalent digital output as the analog input varies. We should make two remarks. First, as V_{in} increases from zero, D_{out} remains at 000 up to $V_{in} = \Delta$. Then, D_{out} jumps to 001 and remains at this value up to $V_{in} = 2\Delta$, etc. We note that, for example, all of the analog values between $V_j = 3\Delta$ and $V_{j+1} = 4\Delta$ are approximated by $D_{out} = 011$. We say the ADC “quantizes” the input signal. Second, since the input and output of the ADC are not equal at all times, we expect that quantization introduces some type of error (Section 12.2).

We employ the following terminology with respect to Fig. 12.1: (1) each jump is called a “code transition”; (2) the input difference between the consecutive jumps, Δ , is called the analog least-significant bit (LSB);¹ (3) the input values at which the code transitions occur are called “decision thresholds”; (4) the unit output jumps are called the digital LSB; and (5) the maximum input level (the total input range) is called the “full scale,” V_{FS} .

¹This holds for an ideal quantizer and will be revisited in Section 12.3.

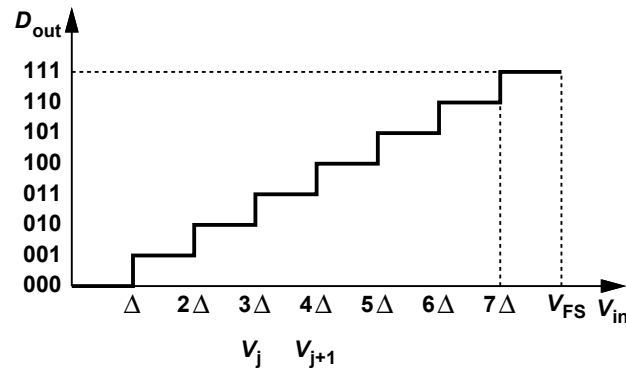


Figure 12.1 A 3-bit ADC characteristic.

12.2 A First Look at Quantization

12.2.1 Basic Concepts

As mentioned in the previous section, we expect that analog-to-digital conversion introduces an error because a quantizer's output is not identical to its input. The phenomenon of quantization can be studied from different perspectives so as to gain interesting insights. We pursue one here and another in Chapter 21.

Let us assume a quantizer characterized by the profile shown in Fig. 12.1. We apply a ramp to the input and examine the output [Fig. 12.2(a)], using the plots in Fig. 12.2(b) to see how V_{in} forms D_{out} . (For ease

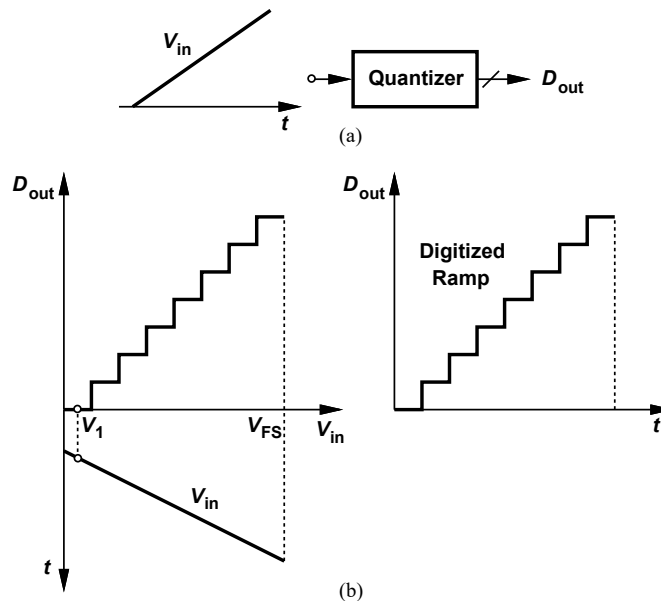


Figure 12.2 (a) A quantizer sensing a ramp, and (b) its output waveform.

of illustration, we plot D_{out} as a continuous-time waveform, as if the ADC were followed by a DAC and we plotted the DAC output.) This arrangement depicts V_{in} versus time, D_{out} versus V_{in} , and D_{out} versus time. As V_{in} reaches V_1 ($= \Delta = 1$ LSB), D_{out} jumps from 000 to 001. This behavior continues for the subsequent codes until V_{in} reaches the full-scale value, V_{FS} . The analog input therefore maps to a “digitized” ramp.

We now loosely define the “quantization error,” ϵ_q , as the difference between the input and output of the quantizer. Superimposing the two as shown in Fig. 12.3(a), and subtracting D_{out} from V_{in} , we obtain ϵ_q as a function of time [Fig. 12.3(b)]. Note that ϵ_q varies between zero and Δ (the analog LSB).

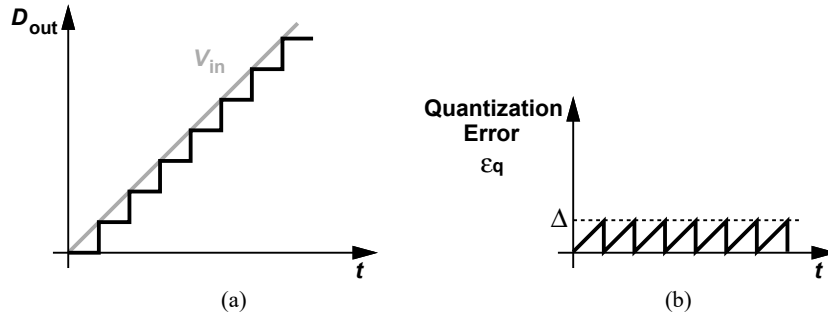


Figure 12.3 (a) Input and outputs waveforms of a quantizer, and (b) the difference between the two.

Example 12.1

For the situation illustrated in Fig. 12.2(a), plot ϵ_q as a function of V_{in} .

Solution

We first plot both D_{out} and V_{in} as a function of V_{in} and then construct $V_{in} - D_{out}$ (Fig. 12.4). The resulting profile suggests that ϵ_q begins at zero for $V_{in} = 0$ and rises linearly as V_{in} increases (because D_{out} remains at zero). At $V_{in} = \Delta$, D_{out} jumps by 1 LSB and $V_{in} = D_{out}$, causing $V_{in} - D_{out}$ to drop to zero, etc. It is important not to confuse the plots in Figs. 12.3(b) and 12.4(b).

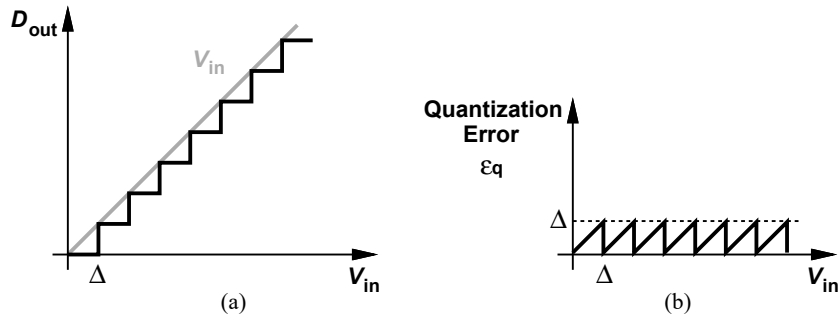


Figure 12.4 Input-output characteristic and quantization error of a quantizer.

The foregoing example teaches two important points. First, since the behavior of ϵ_q versus V_{in} is similar to that of ϵ_q versus time in a ramp test, we can construct the former from the latter. Second, the behavior of ϵ_q versus V_{in} depends on only the level of V_{in} and is independent of the shape of the input in the time domain—just as is D_{out} versus V_{in} .² For example, a sinusoidal V_{in} still generates ϵ_q according to Fig. 12.4, but, of course, a different ϵ_q waveform versus time (Problem 12.1).

12.2.2 Formulation of Quantization Error

We wish to analyze how quantization affects the performance of an ADC. Returning to Figs. 12.3(b) and 12.4(b), we recognize that $0 \leq \epsilon_q \leq \Delta$, i.e., the smaller the analog LSB is, the less ϵ_q is. But we also note that both ϵ_q plots exhibit an average value of $\Delta/2$; this dc component per se does not corrupt the analog signal and is benign in most cases. We then subtract $\Delta/2$ from ϵ_q (Fig. 12.5), observing that the result varies between $-\Delta/2$ and $+\Delta/2$ regardless of the shape of the input waveform. For example, if V_{in} is a random

²This holds only at low input frequencies.

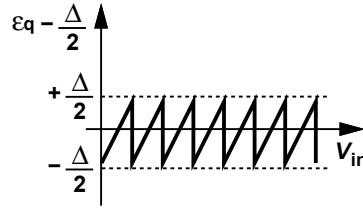


Figure 12.5 Quantization error minus dc offset.

signal, $\epsilon_q - \Delta/2$ fluctuates between those two levels randomly, thereby introducing noise in the digitized waveform. That is, quantization adds noise when the input is random.³ We call $\epsilon_q - \Delta/2$ the “quantization noise” and denote it by n_q .

How do we compute the power of n_q ? We can examine the spectrum of the noise (Chapter 21) or utilize our knowledge of random variables. We know that the “power” of a random variable, x , is given by

$$P_x = \int_{-\infty}^{+\infty} x^2 f(x) dx, \quad (12.3)$$

where $f(x)$ denotes the probability density function (PDF) of x .⁴ For example, a zero-mean Gaussian distribution with a standard deviation of σ carries a power of σ^2 ; hence the useful rule that the rms value of Gaussian noise is equal to its σ . We can apply Eq. (12.3) to quantization noise if $f(x)$ is known. Figure 12.5 suggests that $n_q = \epsilon_q - \Delta/2$ is distributed between $-\Delta/2$ and $+\Delta/2$, but the shape of the distribution for a random input is not known. A simplification that proves relatively accurate in practice is to assume that the quantization noise has a uniform distribution (Fig. 12.6). The height of the PDF is equal to $1/\Delta$ so that the

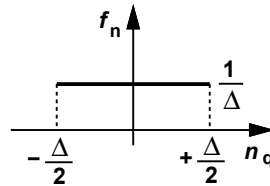


Figure 12.6 Uniform distribution assumed for quantization error.

total area amounts to unity. We can then write from (12.3)

$$P_{nq} = \int_{-\Delta/2}^{+\Delta/2} n_q^2 \frac{1}{\Delta} dn_q \quad (12.4)$$

$$= \frac{1}{\Delta} \int_{-\Delta/2}^{+\Delta/2} n_q^2 dn_q \quad (12.5)$$

$$= \frac{\Delta^2}{12}. \quad (12.6)$$

This important result signifies the corruption introduced by the quantizer. Note that the unit of P_{nq} is V^2 .

Signal-to-Noise Ratio Equation (12.6) implies that P_{nq} can be reduced by scaling Δ down. But what do we do with the full-scale voltage, V_{FS} ? If V_{FS} is maintained constant, then the *number* of quantization steps, m , (the “resolution”) must increase proportionally. Figure 12.7 depicts an example of doubling

³We defer the case of periodic inputs to Chapter 21.

⁴We can view this expression as the average value of x^2 .

the resolution. We expect that the output signal-to-noise ratio (SNR) rises with m . For an N -bit quantizer,

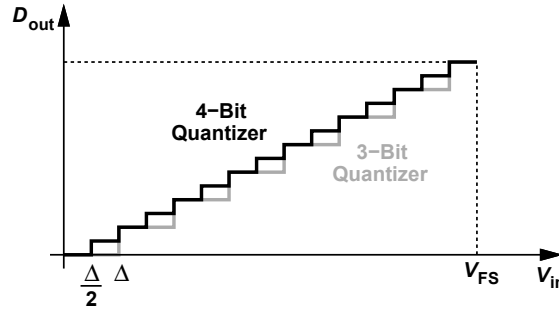


Figure 12.7 Effect of doubling an ADC's resolution.

we have $m = 2^N$. To verify, suppose the full-scale input applied to an N -bit quantizer is expressed as $V_{in} = V_0 \cos \omega_0 t + V_0$, where $V_0 = V_{FS}/2$ (Fig. 12.8). The signal power is given by $V_0^2/2$ and the analog LSB by $\Delta = V_{FS}/2^N = V_0/2^{N-1}$. It follows that

$$\text{SNR} = \frac{\frac{V_0^2}{2}}{\frac{\Delta^2}{12}} \quad (12.7)$$

$$= \frac{3}{2} 2^{2N}. \quad (12.8)$$

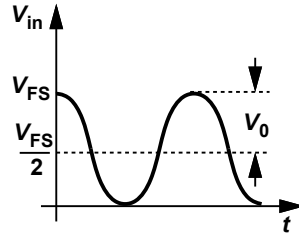


Figure 12.8 Full-scale sinusoid applied to a quantizer.

Taking 10 log of both sides, we have

$$\text{SNR(dB)} = 20N \log 2 + 1.76 \quad (12.9)$$

$$= 6.02N + 1.76. \quad (12.10)$$

Thus, doubling the resolution (i.e., increasing N by 1) raises the SNR by about 6 dB.

Example 12.2

The derivation leading to Eq. (12.10) has assumed that n_q is random and has a uniform distribution. Is this true for a sinusoidal input?

Solution

Strictly speaking, no, it is not. We surmise that a periodic input leads to a periodic output and hence a periodic quantization error. We return to this point in Chapter 21. Nevertheless, (12.6) and (12.10) provide reasonable approximations even for periodic inputs if N is greater than roughly 3.

Equation (12.10) suggests that the SNR is maximized by raising V_0 . Of course, if the input swing exceeds the full scale, the ADC simply generates an output of the form $11 \cdots 1$, i.e., it saturates, introducing high distortion.

12.3 ADC Performance Parameters

In our study of sampling circuits and DACs, we have seen that their performance is quantified by means of numerous parameters. In this section, we present similar parameters for ADCs. The reader is encouraged to first review those of DACs (Chapter 8).

12.3.1 Offset

Consider the simple ADC shown in Fig. 12.9(a). Due to switch charge injection and clock feedthrough, V_X

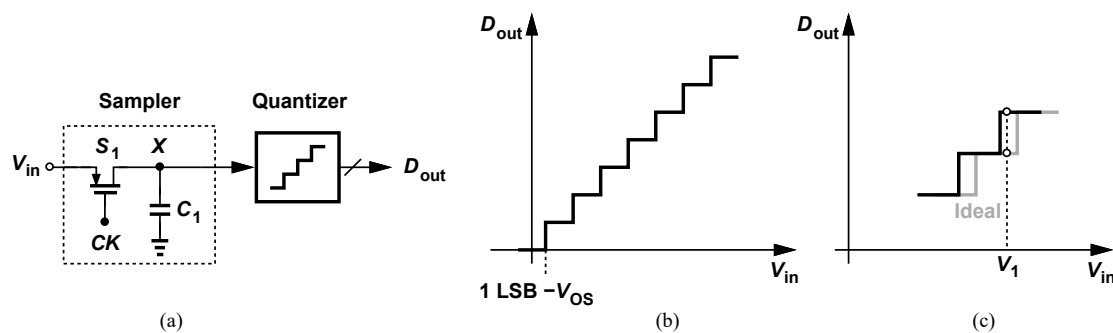


Figure 12.9 (a) A sampler followed by a quantizer, (b) effect of sampler offset on overall characteristic and (c) closeup view of offset.

experiences a positive pedestal equal to V_{OS} when S_1 turns off. That is, if $V_{in} = 0$, $V_X > 0$. More precisely, the first decision threshold of the quantizer is shifted to the left, as illustrated in Fig. 12.9(b). We say the ADC suffers from an offset. The nonidealities within the quantizer itself can introduce offset as well.

It is possible to view the effect of offset as a *vertical* shift. Turning to Fig. 12.9(c), we observe that the digital output corresponding to $V_{in} = V_1$ is *greater* than the ideal case, signifying an upward shift.

12.3.2 Gain Error

Gain error is defined as the departure of the characteristic's slope from its ideal value. Depicted in Fig. 12.10, this phenomenon arises because the analog LSB is greater or less than the nominal. In other words, if $\Delta_2 < \Delta_1$, the gain error is positive and vice versa. We can also see that the full-scale voltage changes. In fact, the *cause* of gain error can be a change in V_{FS} as we have $\Delta = V_{FS}/2^N$.

Example 12.3

A digital thermometer displays temperatures ranging from -20°C to $+60^\circ \text{C}$ with a resolution of 1° . If the ADC used therein exhibits offset and gain error, explain how the temperature readings are affected.

Solution

The ADC offset simply shifts all of the readings up or down by a constant amount. For example, the output values are 2°C greater than the actual temperature.

The ADC gain error multiplies each temperature by a constant coefficient. For example, a gain error of $+2\%$ yields an output equal to 1.02 times the actual temperature. We can see that the reading error *grows* as the temperature reaches the full scale.

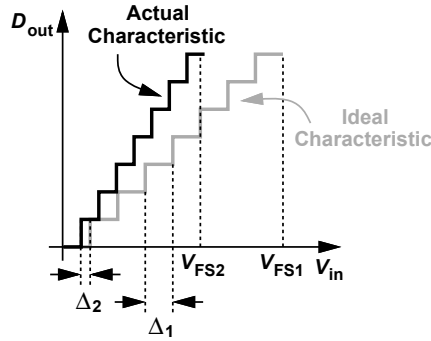


Figure 12.10 Illustration of gain error.

Given an ADC input-output characteristic, how do we compute the gain error? This is not possible if we do not know the ideal analog LSB or the ideal full scale. In contrast to offset, gain error is a *relative* quantity and can only be determined if both characteristics in Fig. 12.10 are known.

12.3.3 Differential Nonlinearity

Our treatment of quantization in the previous sections has assumed evenly-spaced code transitions. But is that true in practice? We must then ask, what type of circuit creates these transitions? As explained in the subsequent chapters, the decision thresholds are determined by one or more comparators, which inevitably suffer from offsets (Chapter 6). In other words, each transition may depart from its ideal position by a random value after an ADC has been fabricated (Fig. 12.11). That is, the input voltage difference between consecutive transitions may not be equal to Δ .

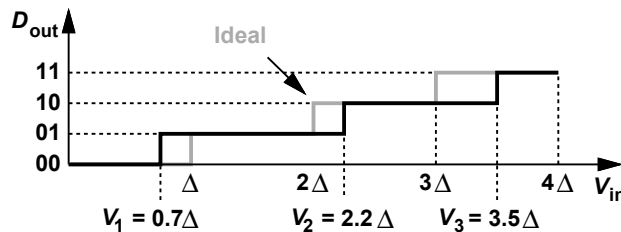


Figure 12.11 DNL in an ADC.

We define the “differential nonlinearity” (DNL) as the deviation in the difference between consecutive transitions from the ideal value of 1 LSB:

$$\text{DNL}_j = V_j - V_{j-1} - 1 \text{ LSB}, \quad (12.11)$$

where V_j and V_{j-1} denote the actual decision thresholds. In Fig. 12.11, for example, we have $\text{DNL}_1 = 0.7\Delta - 0 - \Delta = -0.3 \text{ LSB}$, $\text{DNL}_2 = +0.5 \text{ LSB}$, and $\text{DNL}_3 = +0.3 \text{ LSB}$. Note that there are three DNL values here while the quantizer has four levels.

The reader may wonder how we should plot the DNL: versus V_{in} or versus D_{out} ? In this book, we plot the DNL versus V_{in} as it is more intuitive. In practice, however, it is plotted versus the output “code.” For example, we say the profile shown in Fig. 12.11 exhibits a DNL equal to -0.3 LSB for $D_{out} = 00$, $+0.5 \text{ LSB}$ for $D_{out} = 01$, and $+0.3 \text{ LSB}$ for $D_{out} = 10$. This distinction is unimportant if we are only interested in the maximum DNL.

Example 12.4

An ADC exhibits a DNL of -1 LSB . Explain how its input-output characteristic is distorted. Repeat the exercise if $\text{DNL} = +1 \text{ LSB}$. Assume that only one code transition is offset.

Solution

We write $V_j - V_{j-1} - 1 \text{ LSB} = -1 \text{ LSB}$, obtaining $V_j = V_{j-1}$. Thus, the comparator offset is so large that the two transitions coincide. Illustrated in Fig. 12.12(a), the characteristic displays a jump of 2 LSB at $V_{in} = V_{j-1}$, revealing that code D_j cannot be created for any input value. We say the ADC suffers from a “missing” code.

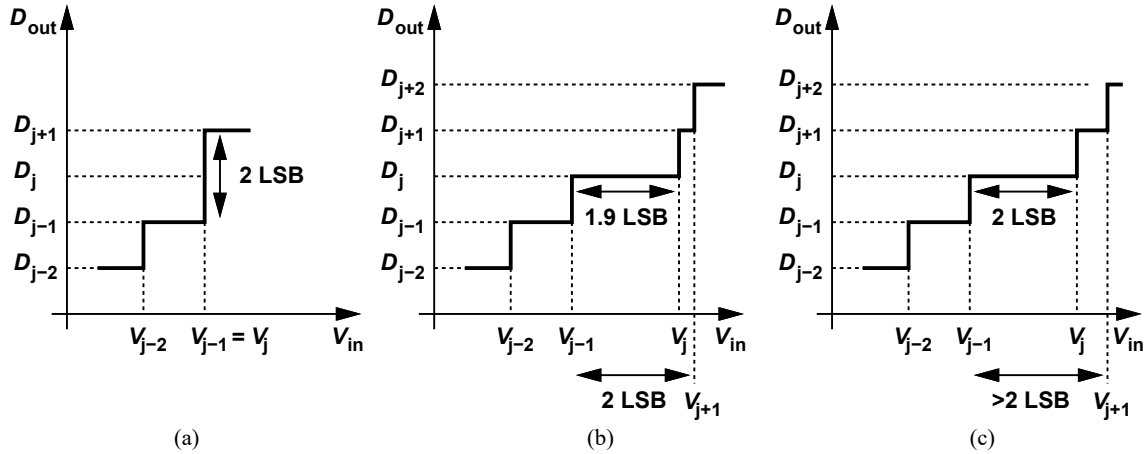


Figure 12.12 (a) Missing code in an ADC, (b) first case of $DNL = +1 \text{ LSB}$, and (c) second case of $DNL = +1 \text{ LSB}$.

To study the case of $DNL = +1 \text{ LSB}$, let us first assume $DNL = +0.9 \text{ LSB}$ [Fig. 12.12(b)]. The corresponding code transitions are 1.9 LSB apart. Now, suppose only the transition at V_j is offset and hence $V_{j+1} - V_{j-1} = 2 \text{ LSB}$; then as DNL_j approaches $+1 \text{ LSB}$, V_j and V_{j+1} coincide, causing code D_{j+1} to vanish.

On the other hand, if both V_j and V_{j+1} are offset and $V_{j+1} - V_{j-1} > 2 \text{ LSB}$, no missing code arises for $DNL_j = +1 \text{ LSB}$ [Fig. 12.12(c)]. In summary, $DNL = -1 \text{ LSB}$ always translates to a missing code but $DNL = +1 \text{ LSB}$ does not.

In ADC design, we target $|DNL| < 1 \text{ LSB}$. In general, differential nonlinearity arises from comparator offset and possibly other circuit imperfections as well. For example, in some ADC architectures, capacitor mismatch leads to DNL. Thus, all DNL contributions must be managed so as to ensure $|DNL| < 1 \text{ LSB}$.

It is important not to confuse the DNL definitions of DACs and ADCs. In the former, the *output* change is considered for DNL computation whereas in the latter, the *input* change is. In other words, the DNL is meaningful for only analog quantities.

Example 12.5

Explain what happens to an ADC's input-output characteristic if the DNL remains positive for part of the input range.

Solution

As shown in Fig. 12.13, the characteristic exhibits a *lower* slope than the ideal value in this range. Conversely, accumulation of negative DNL results in a higher slope.

12.3.4 Integral Nonlinearity

Recall from Chapter 8 that the INL of a DAC represents the curvature in the overall input-output characteristic. We apply the same notion to ADCs. As illustrated in Fig. 12.14(a), we pass a straight line through the

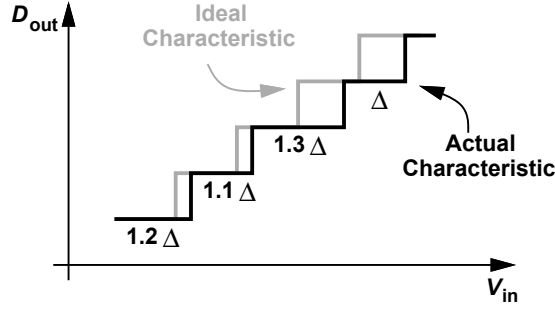


Figure 12.13 A characteristic having $\text{DNL} > 0$ for part of the range.

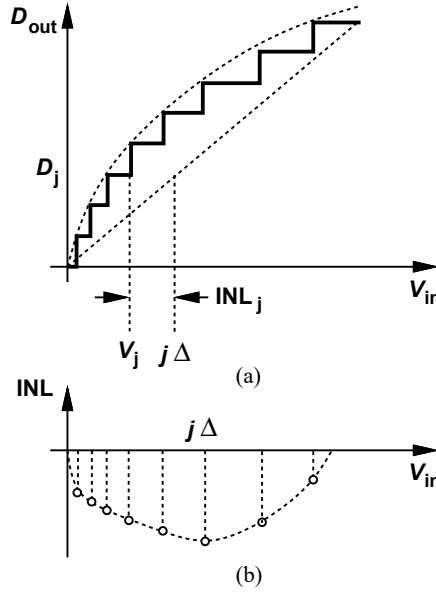


Figure 12.14 (a) Definition of INL in an ADC, and (b) INL profile.

end points and view the *horizontal* difference between the characteristic and this line as the INL profile [Fig. 12.14(b)]. Specifically, if a code transition occurs at V_j rather than at $j\Delta$, we define INL_j as

$$\text{INL}_j = V_j - j\Delta. \quad (12.12)$$

We thus observe that, in both DACs and ADCs, the DNL and the INL are measured along the analog axis. Our comment in the previous section about plotting the profile versus the analog input or versus the output code applies to the INL as well.

Relation Between INL and DNL We have seen in Chapter 8 that the INL of DACs is equal to the integral of their DNL. We then surmise that the same holds for ADCs. Let us write $\text{DNL}_j = V_j - V_{j-1} - \Delta$ and form the summation (integral) thereof as

$$\sum_{k=1}^j \text{DNL}_k = \sum_{k=1}^j (V_k - V_{k-1}) - j\Delta. \quad (12.13)$$

Here, $j\Delta$ represents the straight line in Fig. 12.14(a), i.e., the ideal decision thresholds.⁵ We also have

$$\sum_{k=1}^j (V_k - V_{k-1}) = (V_j - V_{j-1}) + (V_{j-1} - V_{j-2}) + \cdots + (V_1 - V_0), \quad (12.14)$$

where V_0 corresponds to $V_{in} = 0$ (why?). It follows that this summation is simply equal to V_j . Equation (12.13) thus reduces to

$$\sum_{k=1}^j \text{DNL}_k = V_j - j\Delta, \quad (12.15)$$

which is the same as Eq. (12.12).

Example 12.6

The nonlinear characteristic in Fig. 12.14(a) displays a higher slope for low values of V_{in} and a lower slope for high values. Explain how this behavior leads to DNL and INL.

Solution

The higher slope for V_{in} near zero can be viewed as a positive gain error, indicating analog LSB values *less* than the nominal. That is, the DNL is negative and accumulates to yield a negative INL. Beyond the valley INL, the slope of the characteristic falls, the DNL becomes positive, and its accumulation returns the characteristic to the straight line.

Distinction Between INL and Offset The INL profile example shown in Fig. 12.14(b) remains negative for the entire input range. We can argue that this profile, in fact, contains an *offset*, thus overestimating distortion. To appreciate this point, let us shift the profile up so as to create equal and opposite peaks [Fig. 12.15(a)]. This is equivalent to shifting the straight line in Fig. 12.14(a) to the left by the same amount [Fig. 12.15(b)] (why?). We observe that the ADC input-output characteristic introduces two artifacts in the digitized signal: an offset equal to α and a peak INL of $\pm\alpha$.⁶ For convenience, we still pass the straight line through the end points in this book.

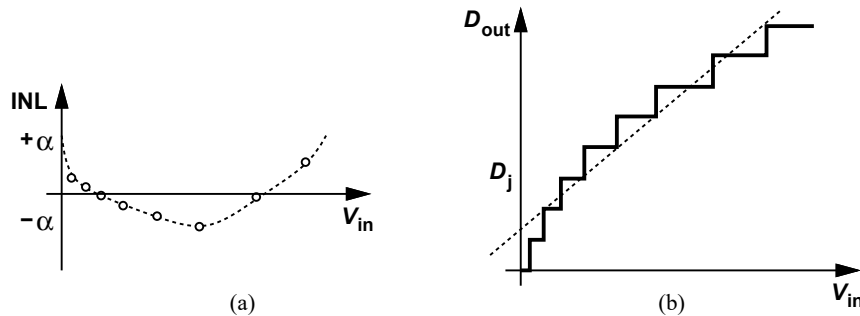


Figure 12.15 (a) INL profile with dc offset removed, and (b) corresponding straight line passing through the characteristic.

INL versus Input Swing We have thus far assumed that the analog input reaches the ADC full-scale voltage. But suppose we apply a smaller input swing, e.g., equal to $V_{FS}/2$ [Fig. 12.16(a)]. Does the signal experience less nonlinearity? Passing a straight line between A and C and finding the difference, we arrive at the two INL

⁵We ignore the effect of offset and gain error here.

⁶Of course, we also have DNL.

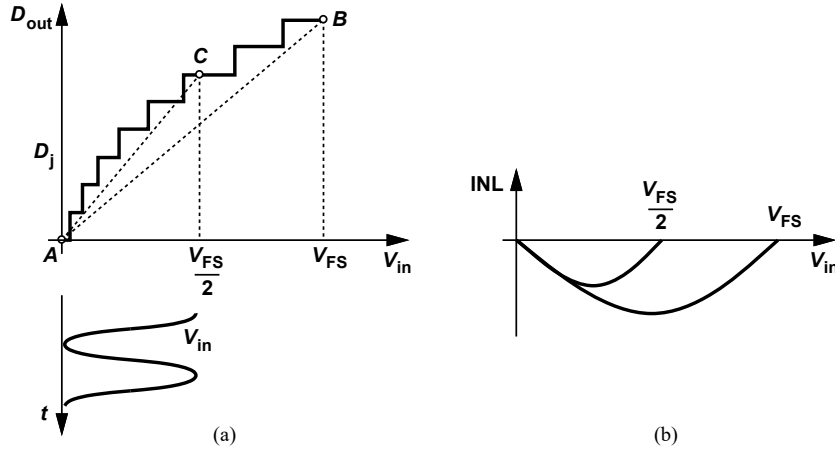


Figure 12.16 (a) An ADC sensing a half-scale input, and (b) reduction of INL.

profiles depicted in Fig. 12.16(b). Thus, the nonlinearity falls for a reduced swing. Unfortunately, however, the signal-to-noise ratio also drops, as predicted by Eq. (12.7). That is, the ADC faces a trade-off between linearity and SNR.

12.3.5 SNDR and ENOB

Recall from Section 12.2.2 that quantization produces noise, yielding an SNR equal to $(3/2)2^{2N}$. In the presence of nonlinearity, i.e., the curvature in Fig. 12.14(a), the ADC distorts the signal as well. We quantify this effect by means of the “signal-to-(noise+distortion)-ratio” (SNDR). For a sinusoidal input with a peak amplitude of V_0 , we have

$$\text{SNDR} = \frac{\frac{V_0^2}{2}}{P_{\text{noise}} + P_{\text{dist}}}, \quad (12.16)$$

where P_{noise} is the total noise power and P_{dist} the total distortion power. The former includes both quantization and electronic noise (e.g., kT/C noise). The latter is obtained by adding the powers of the harmonics that are present in the output. In this section, the term “harmonic” refers to those beyond the fundamental.

Example 12.7

Upon taking the fast Fourier transform (FFT) of an ADC’s output, a student obtains the spectrum shown in Fig. 12.17. Explain how the SNDR is computed from this plot.

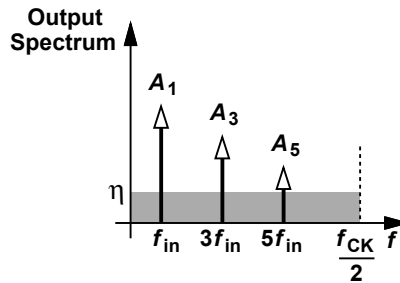


Figure 12.17 An ADC output spectrum.

Solution

We first exclude the impulses at integer multiples of f_{in} and integrate the noise power from $f = 0$ to

$f = f_{CK}/2$.⁷ This yields $P_{noise} = \eta(f_{CK}/2)$ in (12.16). Next, we add the harmonic powers as $P_{dist} = A_3^2/2 + A_5^2/2$, where A_j denotes the *peak* amplitude of harmonic number j . Finally, we substitute these values in (12.16) with $V_0^2/2 = A_1^2/2$.

Example 12.8

An ADC operates near the Nyquist rate, i.e., it senses an input $V_{in} = V_0 \cos \omega_{in} t$, where $\omega_{in} = 2\pi f_{in}$ and $f_{in} = f_{CK}/2 - \Delta f$. How do harmonics appear in the output spectrum?

Solution

Since the harmonics land beyond $f_{CK}/2$, the sampling operation in the ADC aliases them to the first Nyquist zone. Specifically, the second harmonic is convolved (mixed) with the clock frequency, yielding two components at $f_{CK} \pm 2f_{in}$ [Fig. 12.18(a)]. Of interest to us is $f_{CK} - 2f_{in} = 2\Delta f$. Figure 12.18(b) illustrates the overall effect.

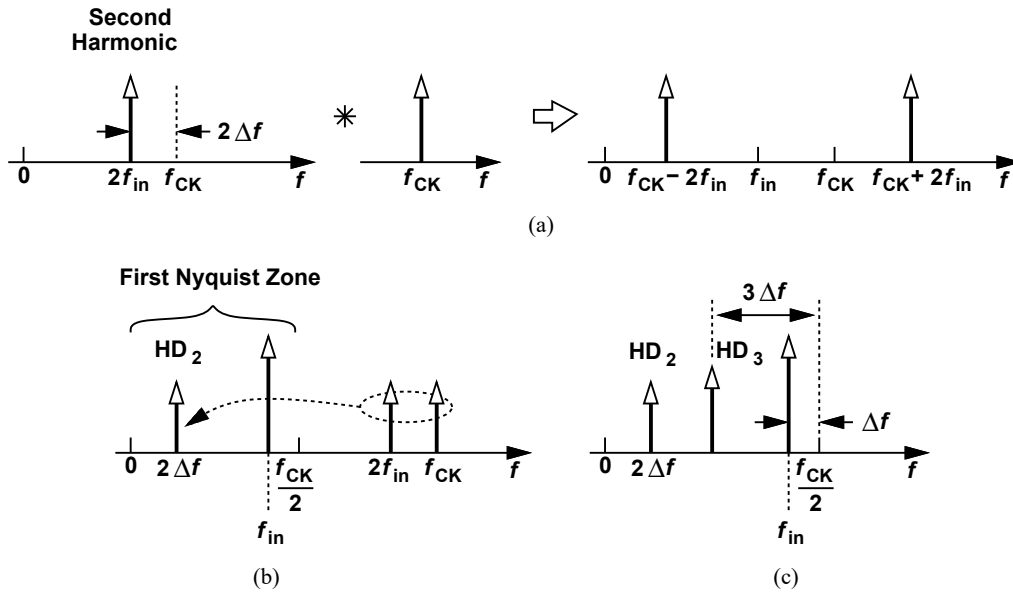


Figure 12.18 (a) Second harmonic generated in an ADC, (b) corresponding folded component, and (c) summary of folded harmonics.

The third input harmonic, given by $3f_{in} = 3f_{CK}/2 - 3\Delta f$, is sampled at a rate of f_{CK} , falling at $f_{CK} - (3f_{CK}/2 - 3\Delta f) = -f_{CK}/2 + 3\Delta f$. This negative value has a counterpart at $f_{CK}/2 - 3\Delta f$. Figure 12.18(c) summarizes the results.

In the presence of electronic noise and distortion, an N -bit ADC's SNDR falls short of the ideal SNR value of $(3/2)2^{2N}$. Equivalently, the ADC no longer provides a resolution of N bits. We can then define an "effective number of bits" (ENOB) so as to quantify the performance; we write (12.10) as

$$\text{SNDR} = 6.02\text{ENOB} + 1.76 \text{ dB} \quad (12.17)$$

⁷Called the "first Nyquist zone," this frequency range is the widest band in which the signal of interest resides.

and hence

$$\text{ENOB} = \frac{\text{SNDR} - 1.76}{6.02}. \quad (12.18)$$

For example, an ADC having a nominal resolution of 10 bits and exhibiting an SNDR of 58 dB provides $\text{ENOB} = 9.3$ bits.

Example 12.9

We evaluate the performance of an ADC by applying a sinusoidal input, gradually increasing its amplitude, and measuring the SNDR. Does the SNDR reach a maximum for an input swing equal to the full scale?

Solution

In typical designs, yes, it does. Equation (12.16) reveals two opposing mechanisms as V_0 increases: both the numerator and P_{dist} rise. Nonetheless, the numerator climbs more steeply up to a certain point. As P_{dist} exceeds P_{noise} , the growth of SNDR slows down.

12.3.6 SFDR

The “spurious-free dynamic range” (SFDR) of an ADC is defined as the ratio of the desired signal power to the highest unwanted component power (Fig. 12.19). Note that the SFDR does not account for noise.

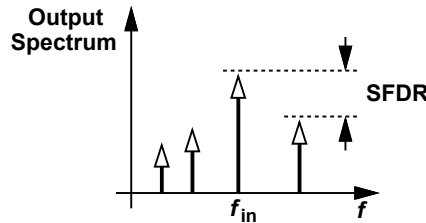


Figure 12.19 Definition of SFDR.

As a measure of ADC’s performance, the SFDR is less rigorous than the SNDR. The former can generally be improved by selecting a moderate or small input swing whereas the latter cannot. To understand this point, recall from Chapter 2 that the amplitude of the n -th harmonic grows in proportion to the n -th power of the input amplitude, V_0 . For example, suppose the third harmonic is the highest unwanted component in Fig. 12.19. If V_0 is halved, this harmonic falls by a factor of 8, raising the SFDR. One must then specify the input swing for which the SFDR is measured.

12.3.7 Static and Dynamic Performance

An N -bit ADC operating at a conversion rate of f_{CK} must ideally provide an SNR and SNDR of $(3/2)2^{2N}$ at all input frequencies below the Nyquist rate. In practice, however, the performance degrades as f_{in} increases. In fact, the SNDR and ENOB may fall considerably as f_{in} approaches $f_{CK}/2$. For this reason, we distinguish between “static” and “dynamic” performance. The former refers to the characteristics measured with a low input frequency, often simply a slow ramp. The latter signifies the ADC behavior with a full-scale sinusoidal input whose frequency is varied from low values to near $f_{CK}/2$.

12.3.8 Figure of Merit

The performance of ADCs is characterized by three metrics, namely, the resolution, the conversion rate, and the power consumption. We can then define a figure of merit (FOM) that encapsulates the trade-offs

among these parameters. Since we expect that doubling the resolution or speed requires doubling the power consumption, we write

$$\text{FOM}_1 = \frac{P}{f_{CK} \cdot 2^N}. \quad (12.19)$$

Note that P/f_{CK} in fact signifies the energy consumed by the ADC in one clock cycle. For this reason, we express FOM_1 in joules per conversion cycle. Of course, we wish to *minimize* FOM_1 .

FOM_1 , however, does not represent the *dynamic* performance of ADCs, failing to show how the ENOB falls as the analog input frequency approaches $f_{CK}/2$. We therefore replace 2^N with 2^{ENOB} and f_{CK} with twice the input frequency at which the ENOB is measured:

$$\text{FOM}_2 = \frac{P}{2f_{in} \cdot 2^{\text{ENOB}}}. \quad (12.20)$$

FOM_2 is expressed in joules per conversion cycle as well. Ideally, $f_{in} \approx f_{CK}/2$ and $\text{ENOB} = N$. But in practice, the minimum FOM_2 may occur below the Nyquist rate, depending on how steeply the ENOB falls versus f_{in} . Today's ADCs achieve FOM_2 values as low as a few tens of femtojoules per conversion cycle.

12.4 Nyquist-Rate ADC Architectures

Extensive work on ADCs has resulted in numerous architectures for Nyquist-rate operation. Depending on the resolution and the speed, we prefer some over others. Illustrated in Fig. 12.20 are some examples. Flash

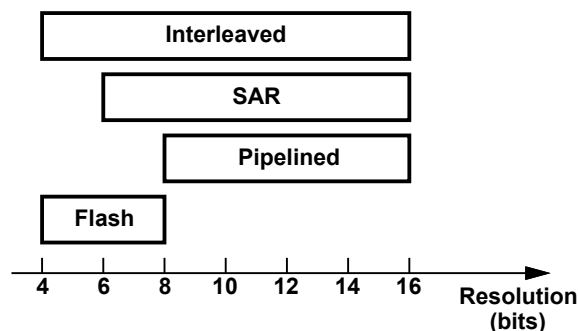


Figure 12.20 Most popular Nyquist-rate ADC architectures.

topologies (Chapter 13) are suited for roughly $4 \leq N \leq 8$. A flash variant employing “folding” and “interpolation” proves useful for $6 \leq N \leq 10$ but is less common. We describe a simple, efficient interpolation scheme in the next chapter. Pipelined ADCs (Chapter 17) find application for $8 \leq N \leq 16$, and successive-approximation (SAR⁸) architectures (Chapter 14) for $6 \leq N \leq 16$. Among these, SAR ADCs are the most popular today for their potentially low power consumption.

The concept of “interleaving” can be applied to any ADC architecture to achieve a higher conversion rate and/or ease the speed-power trade-off (Chapter 20).

⁸SAR traditionally stands for “successive-approximation register,” but we drop the third word here.

Problems

- 12.1. If $V_{in} = V_0 \cos \omega_0 t + V_0$ in Fig. 12.2 and $2V_0 \approx V_{FS}$, sketch the analog equivalent of the quantizer's output as a function of time.
- 12.2. We double the input voltage range of a 10-bit quantizer. Explain what happens to the Q-noise power and the maximum SNR.
- 12.3. A 12-bit quantizer operates with a full-scale voltage of 1 V.
 - (a) Determine the Q-noise power.
 - (b) Determine the output SNR for a full-scale sinusoidal input.
 - (c) How much does the SNR degrade if the sinusoidal amplitude is halved?
- 12.4. If we double the capacitor value in Fig. 12.9(a), what happens to the offset and the gain error?
- 12.5. Consider the 3-bit quantizer in Fig. 12.2(b). Suppose $DNL = 0.2$ LSB for all of the input codes. Sketch the resulting input-output characteristic.
- 12.6. Sketch the INL profile for the previous problem.
- 12.7. In a hypothetical 3-bit ADC, we have $DNL_j = 0.1j$ for $j = 1, \dots, 8$. Compute the gain error and the INL profile.
- 12.8. An ADC sensing an input sinusoid exhibits second and third harmonics that are respectively 40 dB and 50 dB below the fundamental. Determine the SNDR. [Hint: The power of the fundamental is expressed as $A^2/2$, where A is the peak input amplitude, and that of the second harmonic as $(A^2/2)/(10^{-40/10})$.]
- 12.9. An ADC operates near the Nyquist rate with an input of the form $V_{in} = V_0 \cos \omega_{in} t$. The circuit generates the fifth harmonic of the input. Plot the output spectrum.
- 12.10. A 3-bit ADC exhibits an INL profile given by $INL_j = j^2 - 8j$. Sketch the DNL profile.
- 12.11. An ADC exhibits an input-output characteristic given by $y(t) = \alpha_1 x(t) + \alpha_3 x^3(t)$. The input range is from zero to x_{max} . (a) Determine the INL profile and its maximum value. (b) If $x(t) = A \cos \omega_{in} t$, calculate the peak amplitude of the third harmonic and express it in terms of the maximum INL.
- 12.12. In the previous problem, we halve the input amplitude while the input range is still equal to x_{max} . What happens to the peak amplitude of the third harmonic and the Q-noise power?